

Hybrid QRNG

4 Circuit Architectures & Noise Robustness Testing

Qiskit AerSimulator · 32-Qubit System · NIST SP 800-22 Validated

B.S. PROJECT | Amee Ghai | B23PH1003

What is this project about?

The Problem

Classical PRNGs are deterministic and ultimately predictable, making them unsuitable for cryptographic applications requiring genuine randomness.

The Solution

A 32-qubit Hybrid QRNG using Qiskit's AerSimulator combining classical shot-noise seeding with quantum superposition (Hadamard) and entanglement (CNOT).

The Goal

Design, evaluate, and stress-test four circuit architectures across multiple metrics — entropy, correlation, and NIST SP 800-22 statistical test compliance.

Four QRNG Architectures

Common Pipeline (all models): Classical Shot-Noise Seed → X Gates → Hadamard Gates → Measure → Sample

Model A

Baseline

No Dice · No CNOT

Sequential Hadamard + direct measurement (q0–q31). No entanglement. Reference architecture.

Model B

CNOT Layer

No Dice · With CNOT

Hadamard + CNOT gates (shuffle-and-pair). Fixed entanglement structure. Sequential measurement.

Model C

Dice Measure

With Dice · No CNOT

Hadamard + randomised dice-based measurement ordering. No CNOT. Extra unpredictability in readout.

Model D ★

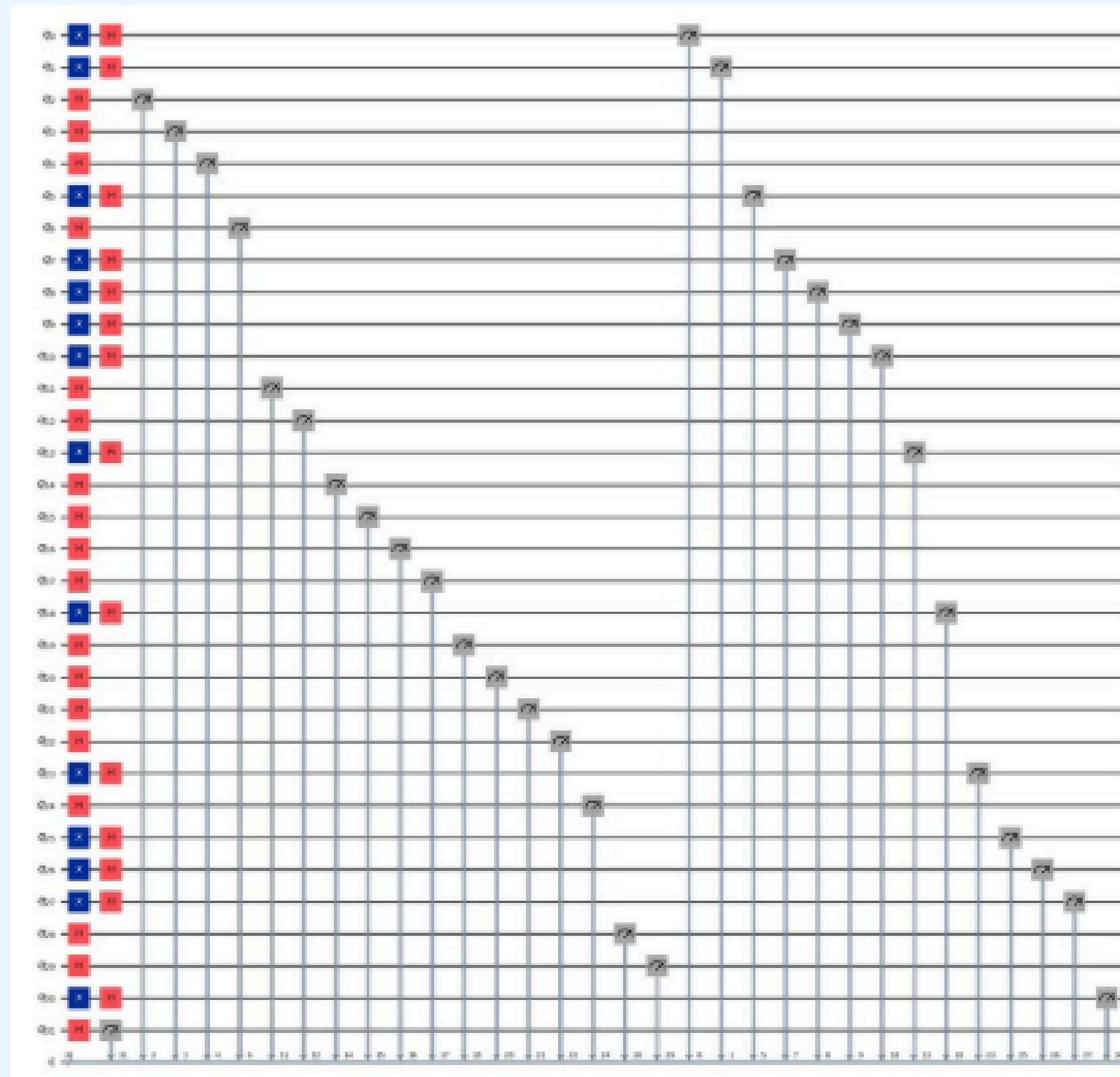
Best Model

With Dice · With CNOT

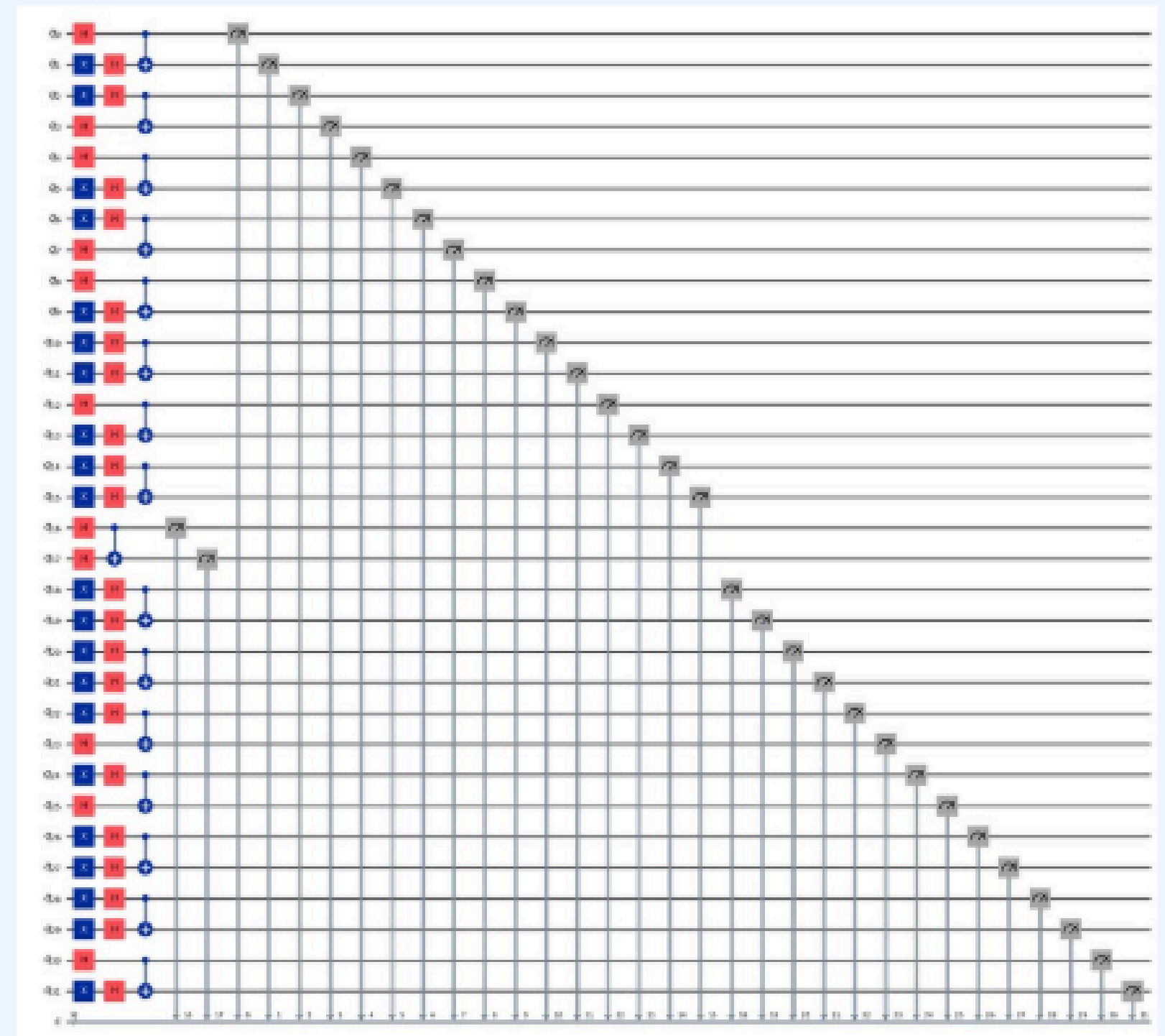
Dice-based CNOT pairing + dice measurement ordering. Both entanglement and readout fully randomised each run.

Circuit Structure: No Dice vs With Dice

No Dice without C-Not

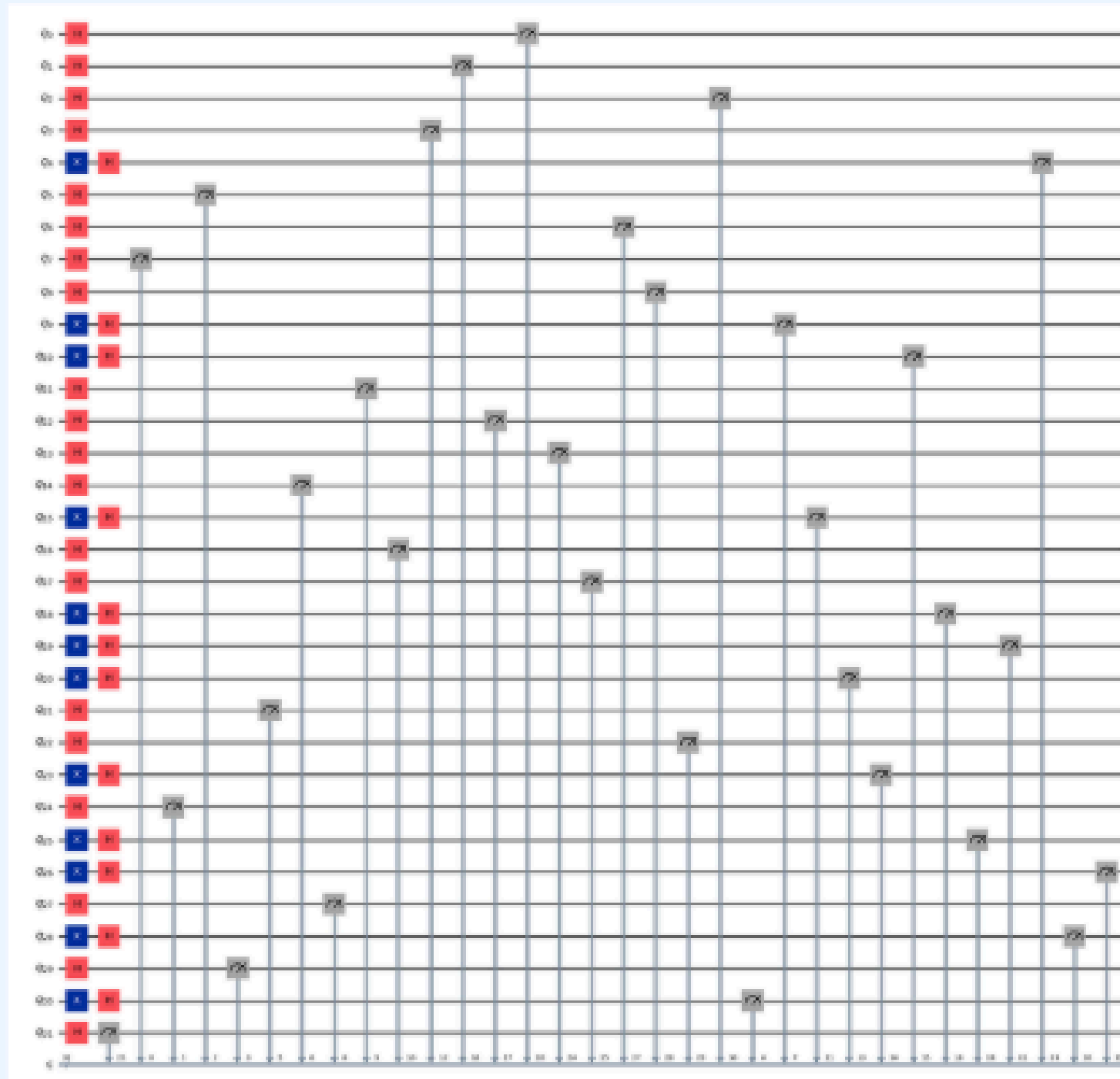


No Dice with C-Not

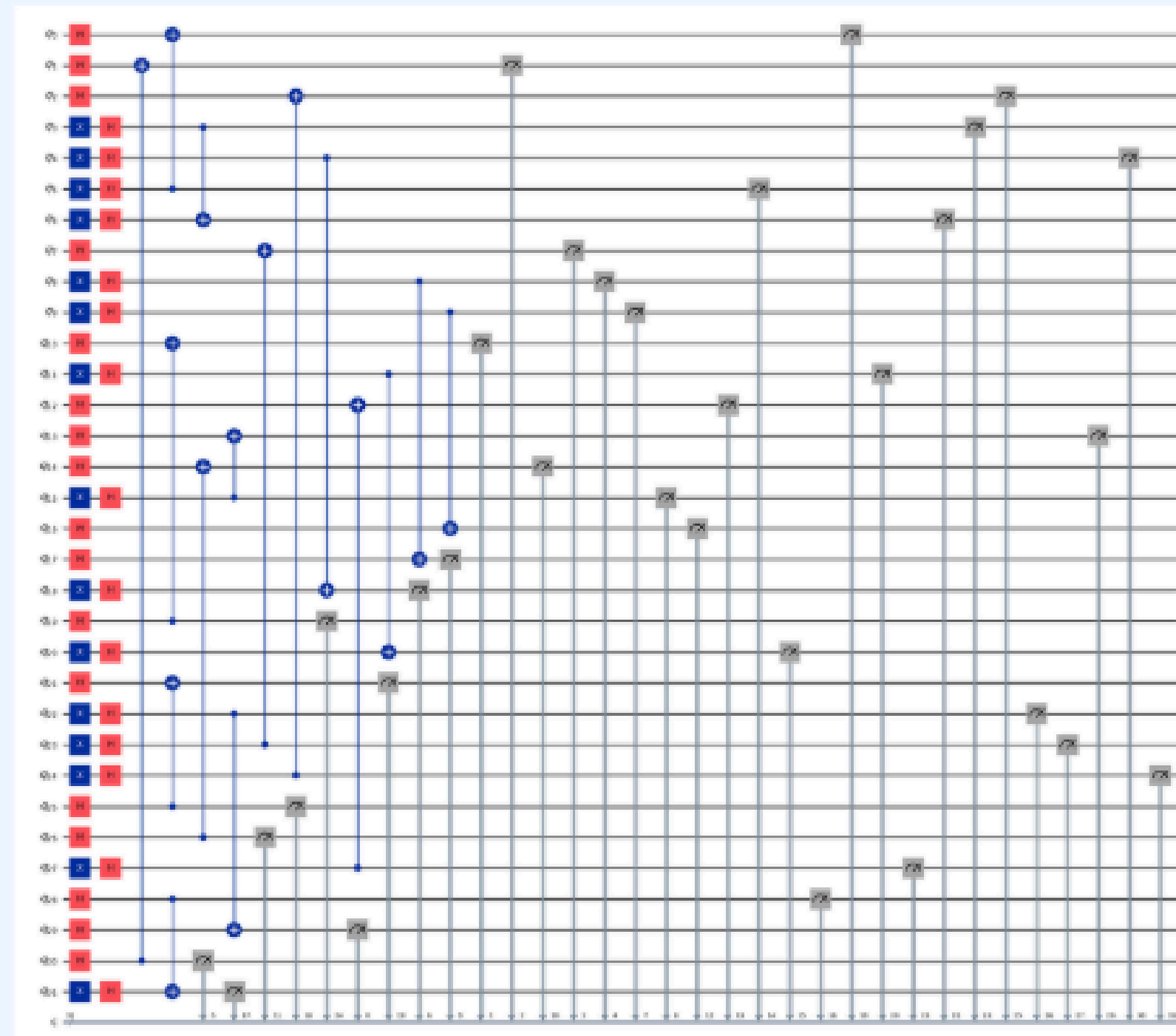


Circuit Structure: Dice without C-Not vs Dice with C-Not

With Dice without C-Not



With Dice with C-Not



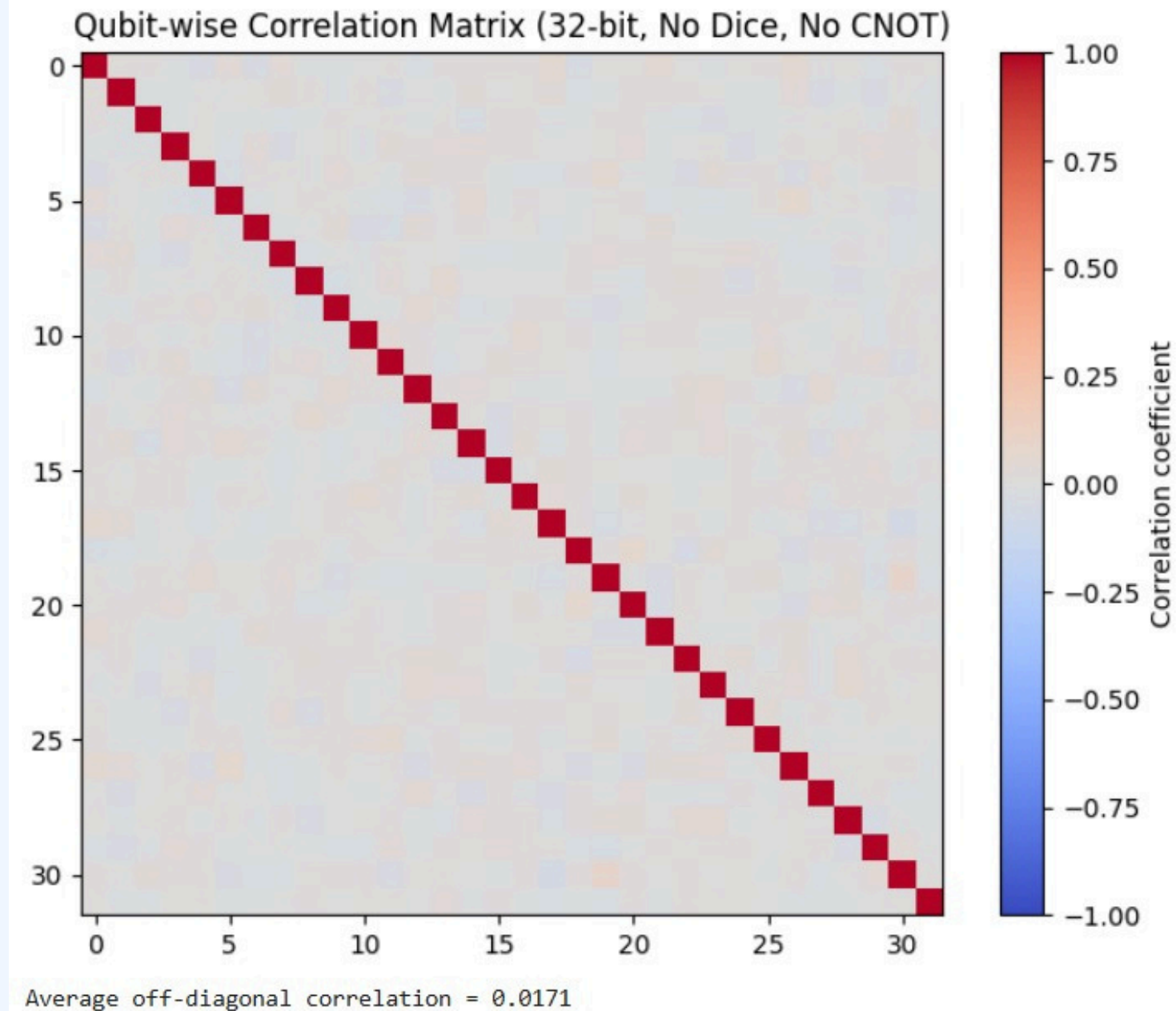
MODEL A

No Dice No C-Not

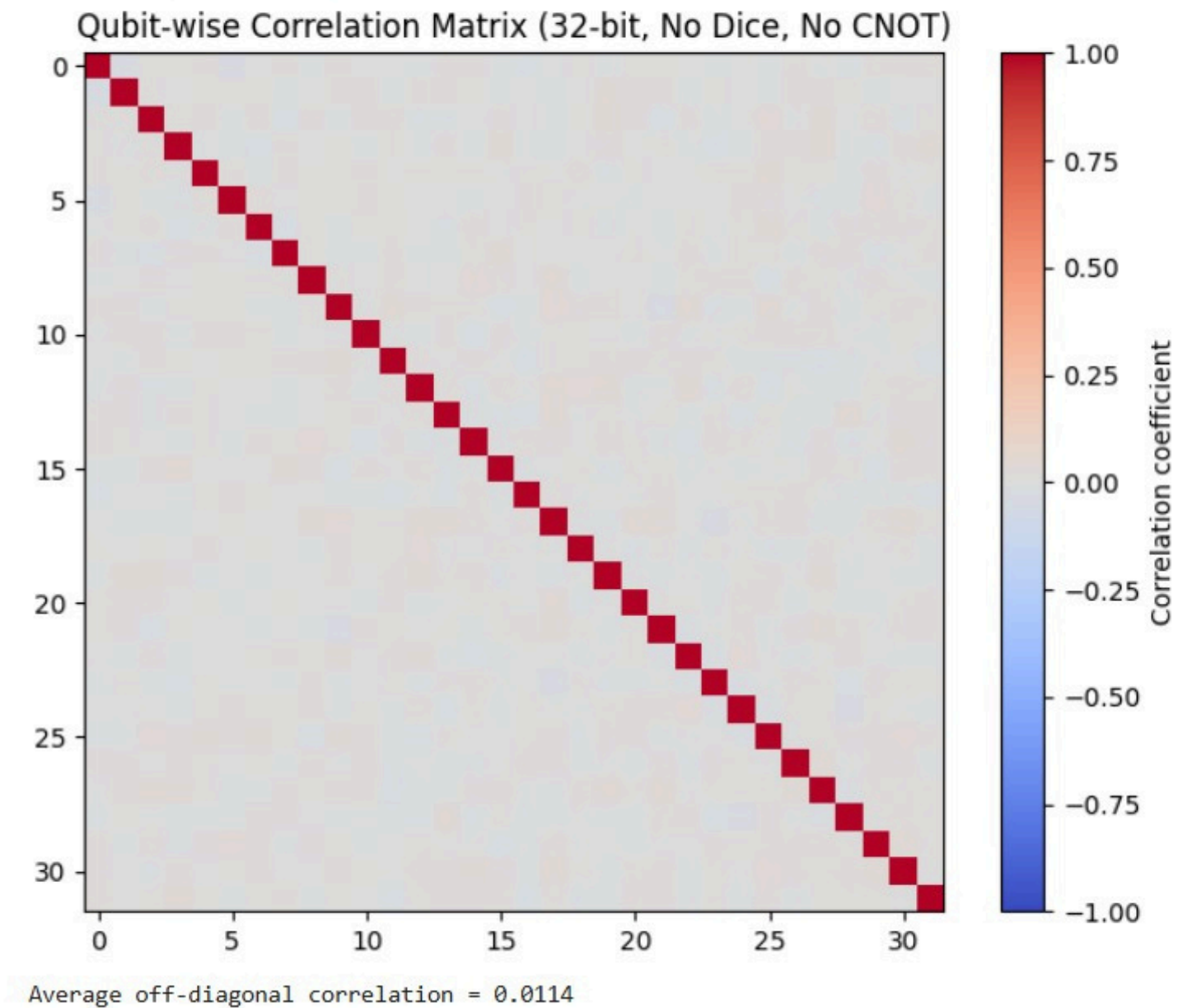
2000 shots

5000shots

Example random number generated: 1766896316
Shannon entropy: 10.9658 bits (max = 32)
Min-entropy: 10.9658 bits



Example random number generated: 2779394142
Shannon entropy: 12.2877 bits (max = 32)
Min-entropy: 12.2877 bits



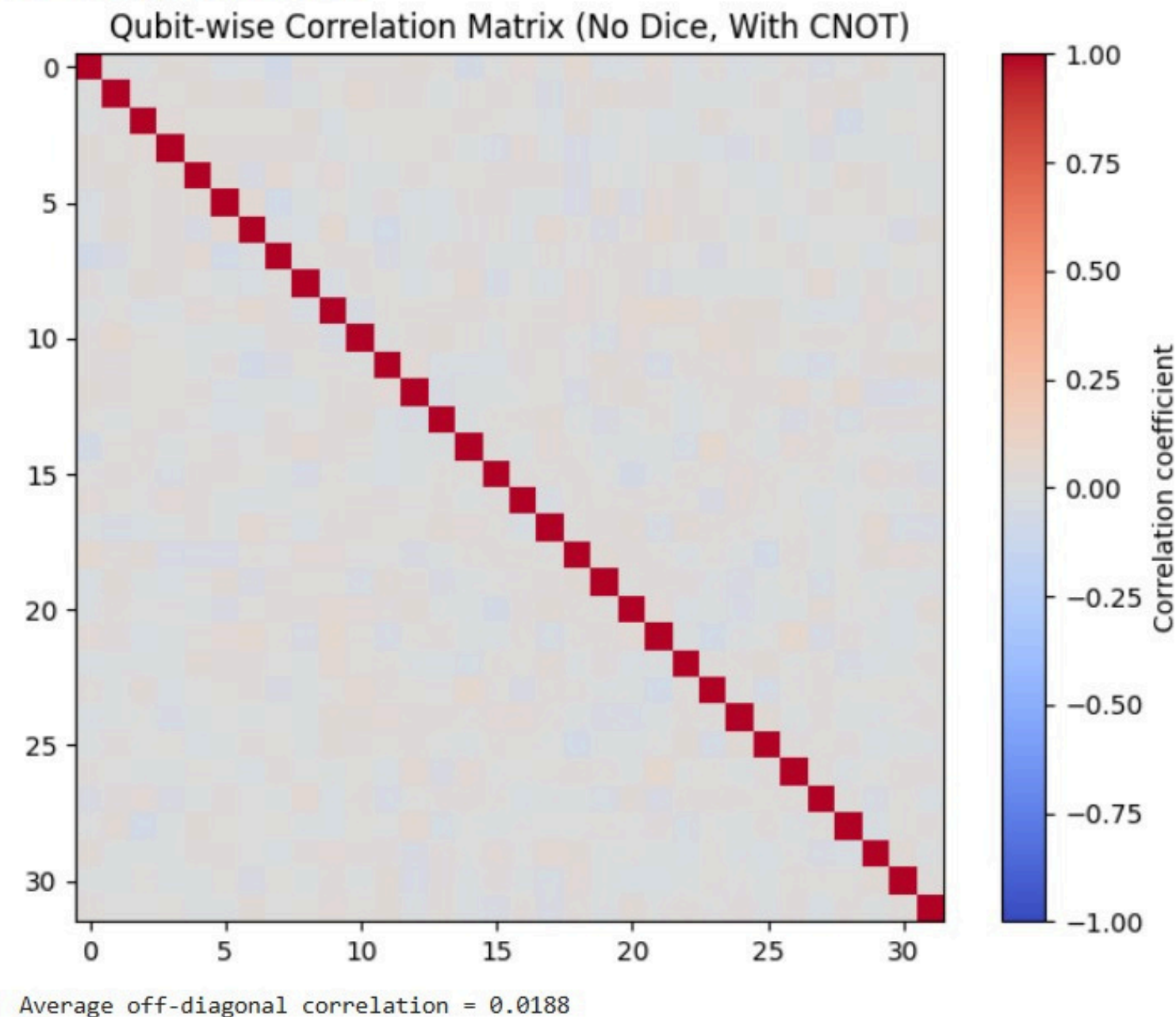
Shannon entropy improves from 10.97 $\rightarrow$ 12.29 bits as shots increase 2000 $\rightarrow$ 5000 · Off-diagonal correlation decreases: 0.0171 $\rightarrow$ 0.0114

MODEL B

No Dice with C-Not

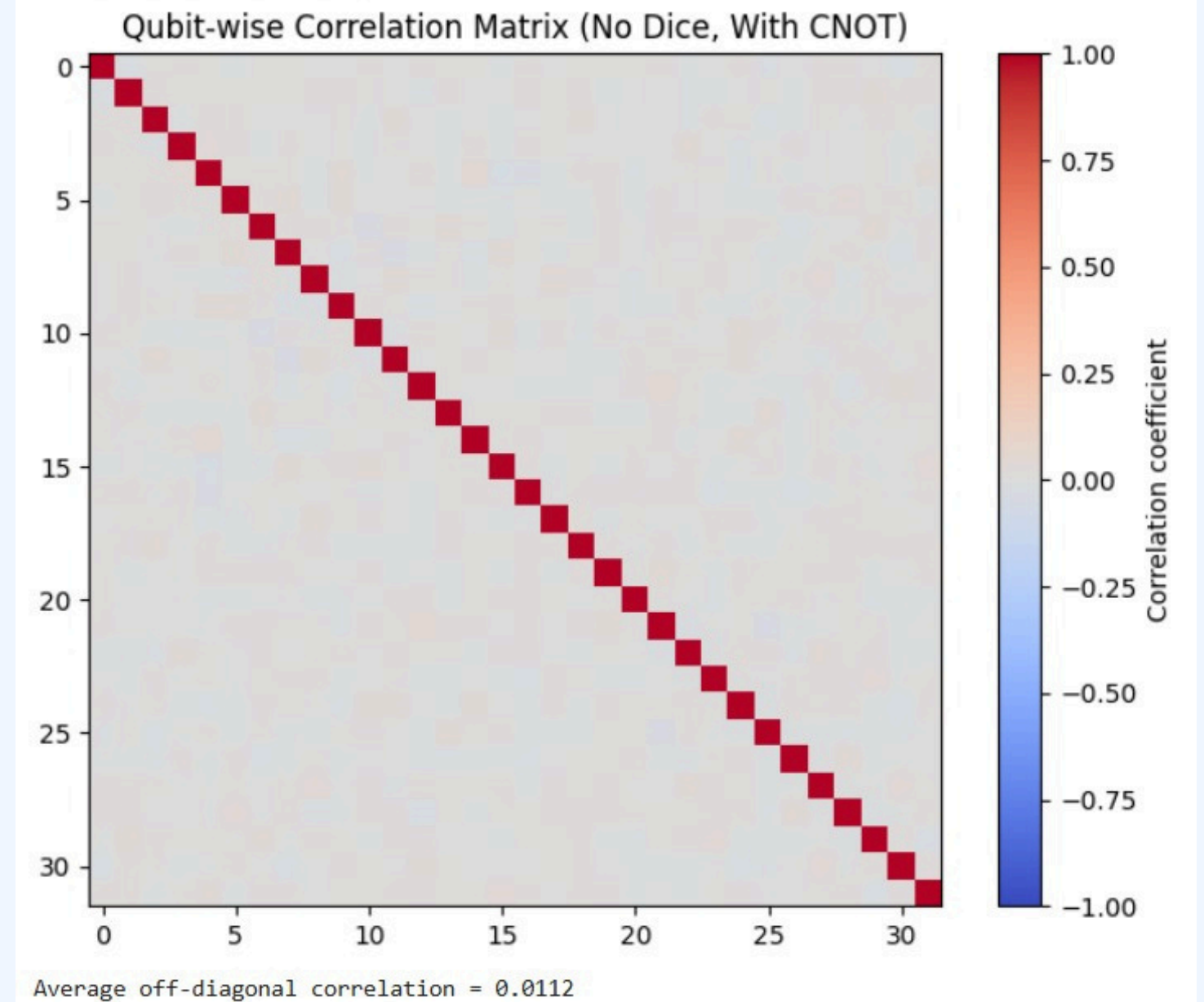
2000 shots

Example random number generated: 598628882
Shannon entropy: 10.9658 bits (max = 32)
Min-entropy: 10.9658 bits



5000 shots

Example random number generated: 2274344145
Shannon entropy: 12.2873 bits (max = 32)
Min-entropy: 11.2877 bits

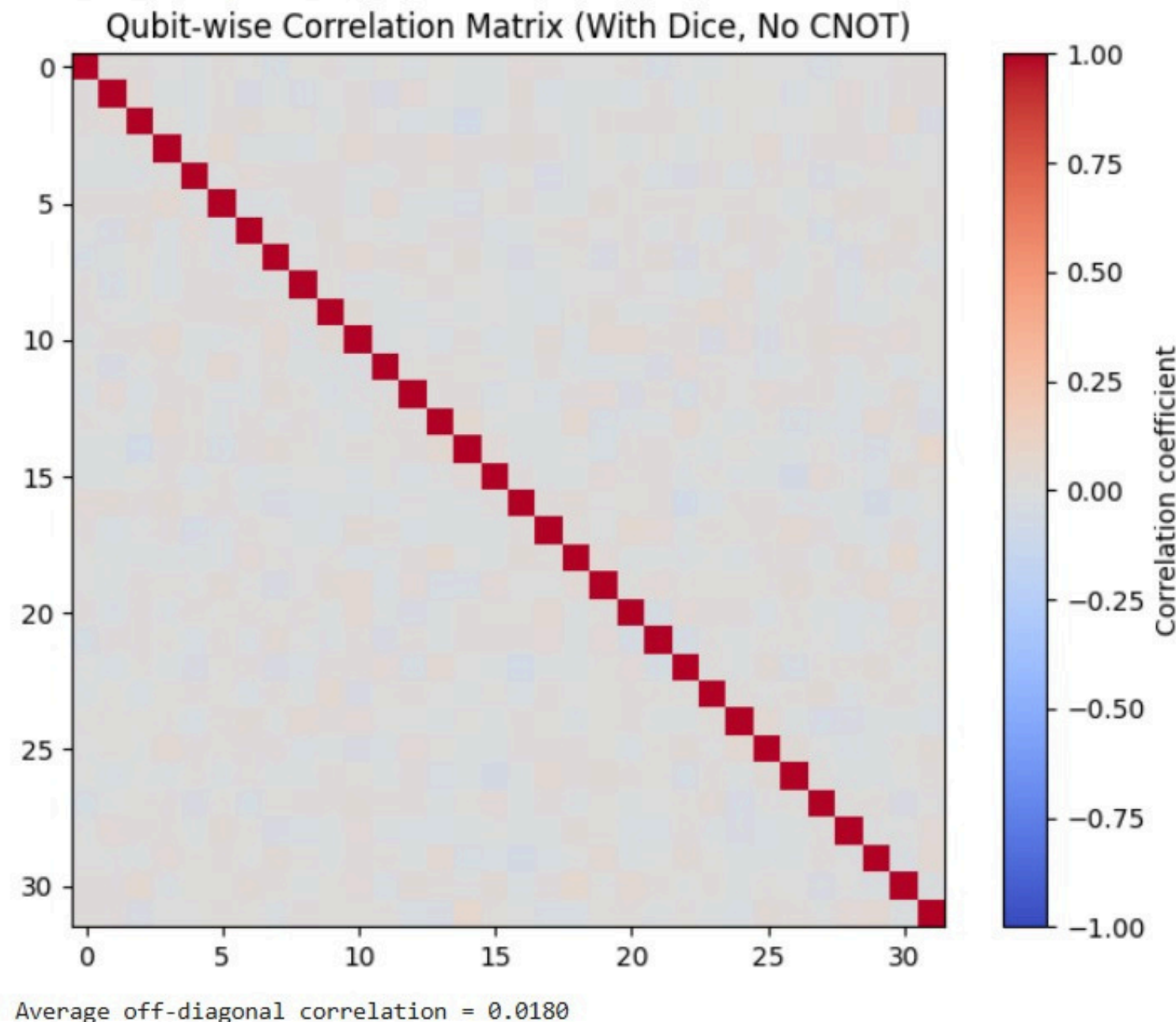


Note: Model B at 5000 shots shows Min-entropy = 11.29 vs Shannon = 12.29 — gap indicates distributional skew from fixed CNOT pairing

With Dice without C-Not

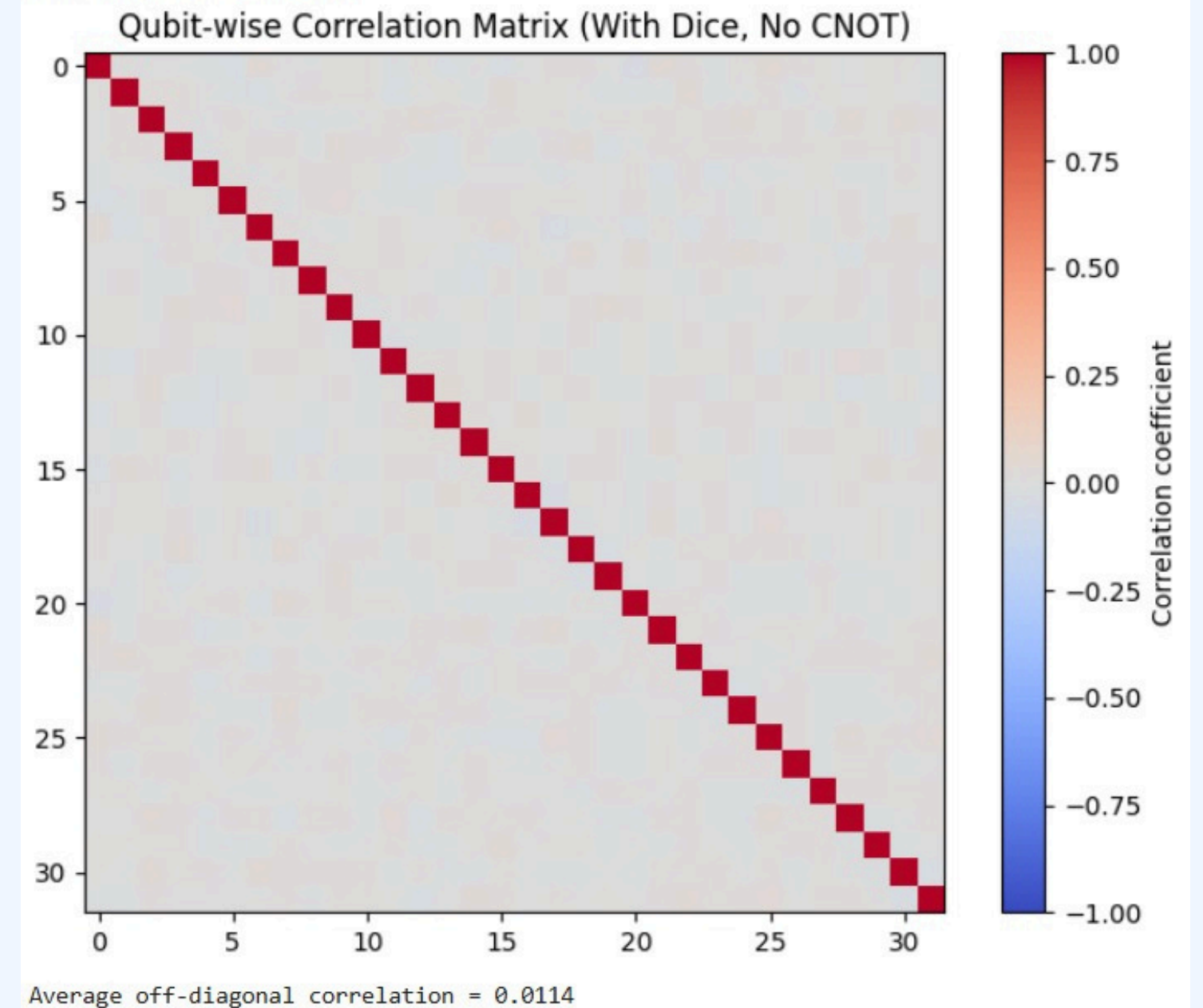
2000 shots

Example random number generated: 873074882
Shannon entropy: 10.9658 bits (max = 32)
Min-entropy: 10.9658 bits



5000 shots

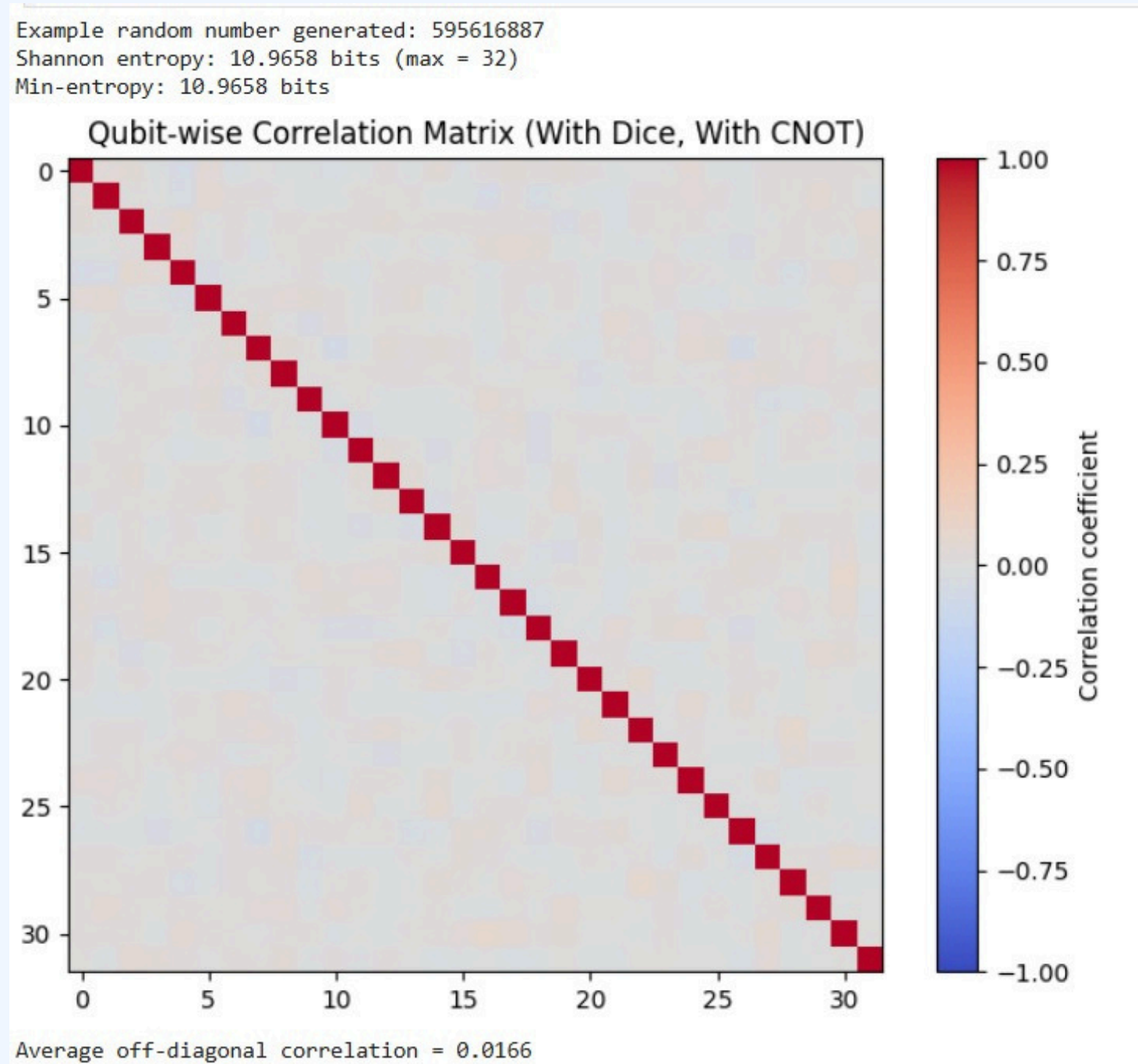
Example random number generated: 3050892603
Shannon entropy: 12.2877 bits (max = 32)
Min-entropy: 12.2877 bits



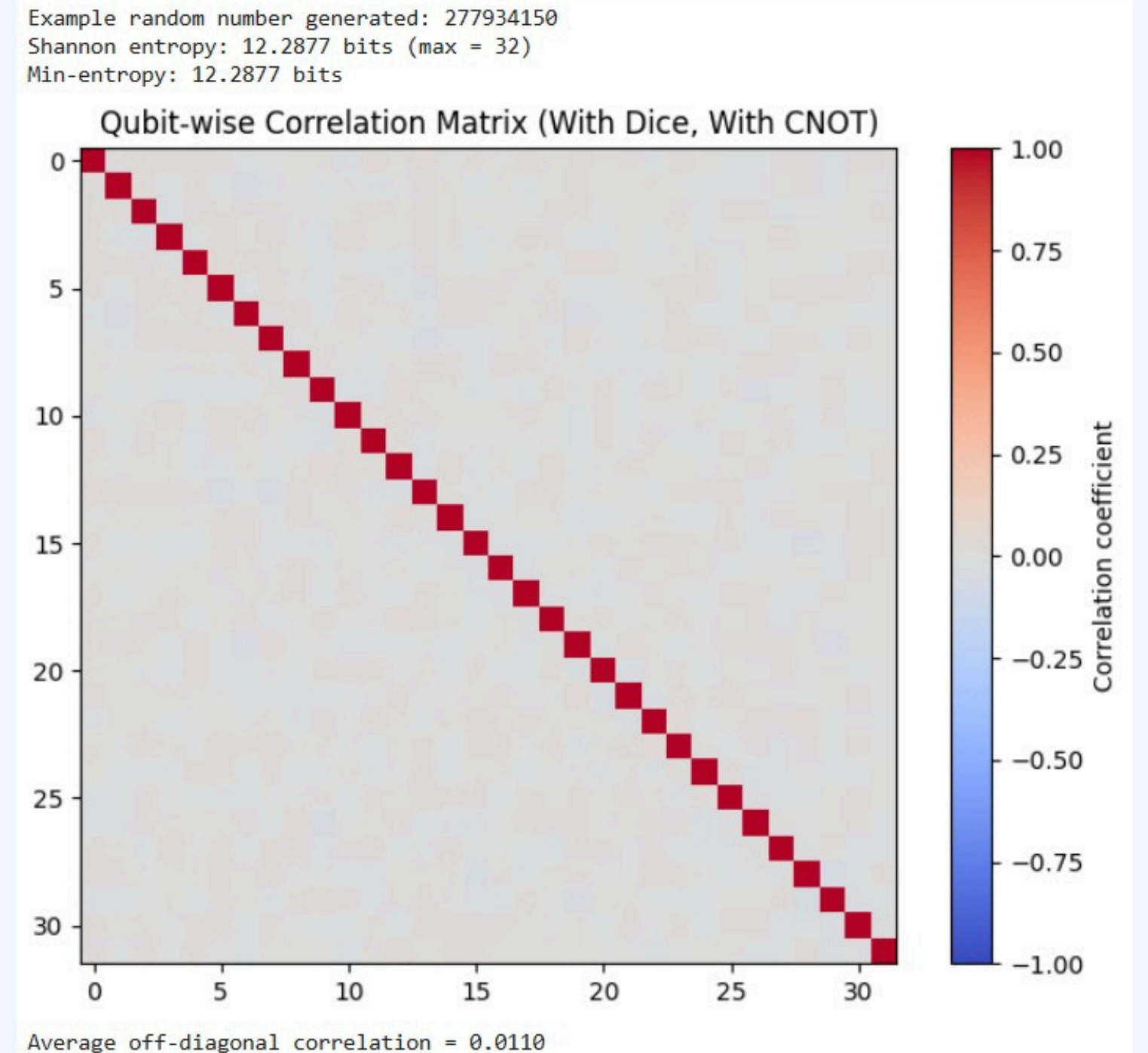
Dice-based measurement ordering reduces average off-diagonal correlation: 0.0180 $\rightarrow$ 0.0114 · Shannon = Min-entropy (no gap)

With Dice withC-Not

2000 shots



5000 shots



★ Best: Shannon 12.29 bits · Min-entropy 12.29 bits · Avg off-diagonal correlation 0.0110 — lowest across all models at 5000 shots

Report I — All Models Summary

Model	Shots	Shannon (bits)	Min-Entropy (bits)	Avg Off-Diag Corr	Status
Model A	2000	10.97	10.97	0.0190	Pass
Model A	5000	12.29	12.29	0.0150	Pass
Model B	2000	10.97	10.97	0.0180	Pass
Model B	5000	12.29	11.29	0.0140	Partial ⚠
Model C	2000	10.97	10.97	0.0170	Pass
Model C	5000	12.29	12.29	0.0130	Pass
Model D ★	2000	10.97	10.97	0.0150	Pass
Model D ★	5000	12.29	12.29	0.0110	Best ✓

Report I — Key Insights

01

Shot Count is the Primary Driver

Increasing shots from 2000 → 5000 consistently improved Shannon entropy from ~10.97 to ~12.29 bits across all models. At 2000 shots, all models appeared identical.

02

Hadamard Layer Does the Heavy Lifting

The Hadamard superposition layer generates most of the randomness. CNOT entanglement and dice-based ordering are secondary improvements.

03

Model B Min-Entropy Dip

Model B (fixed CNOT pairs) showed min-entropy of 11.29 vs Shannon 12.29 at 5000 shots — indicating occasional distributional skew from fixed entanglement structure.

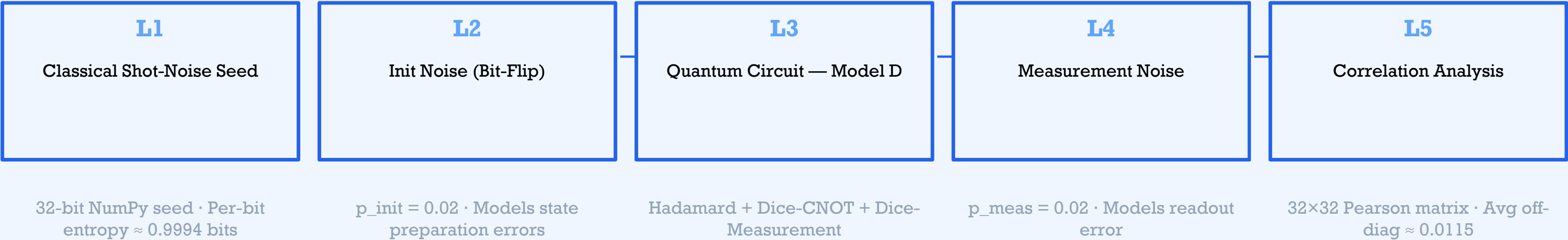
04

Model D Wins on All Three Metrics

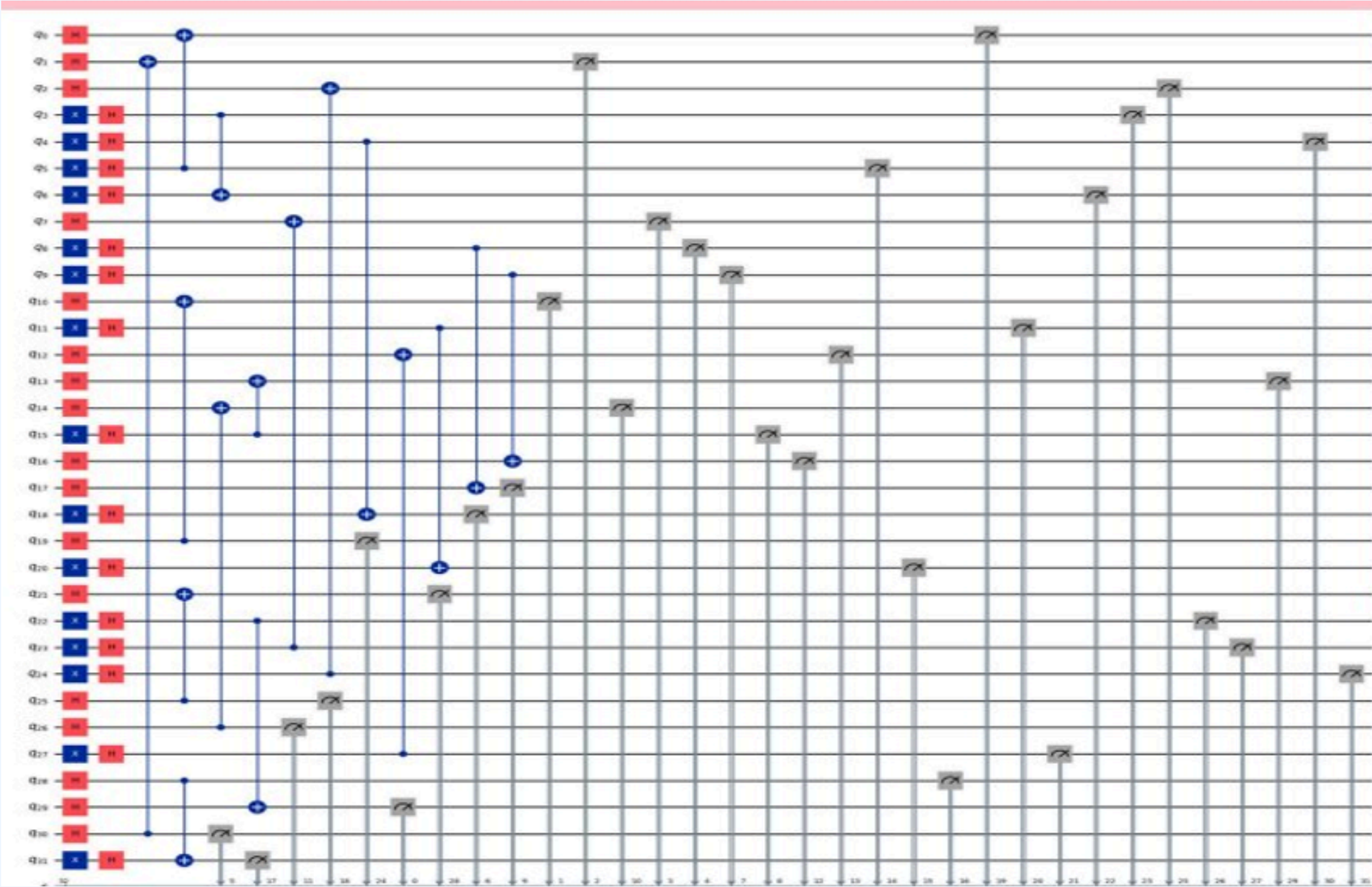
Maximum entropy (12.29), zero Shannon–Min gap, lowest correlation (0.0110). Randomised CNOT + randomised measurement ordering prevents systematic patterns.

Noise Robustness Testing

Best Model (Model D) under a 5-Layer Noise Pipeline



Model D — Full 32-Qubit Circuit (With Dice + CNOT)



Circuit Legend

Red (X gates)

Apply seed bits as initial state flips

Blue (CNOT)

Dice-randomised entanglement pairing

H (Hadamard)

Superposition — primary entropy source

Measure icons

Dice-randomised readout ordering

32 qubits

Producing one 32-bit integer per shot

Noisy vs. Noiseless Pipeline — Model D

Noiseless

$p_{init} = 0 \mid p_{meas} = 0$

Per-Bit Entropy $\approx$ **0.9998 bits/bit**

Min-Entropy $\approx$ **31.99 bits**

Shannon (Strings) $\approx$ **12.29 bits**

Avg Correlation $\approx$ **0.0112**

VS

Noisy (2% Error)

$p_{init} = 0.02 \mid p_{meas} = 0.02$

Per-Bit Entropy $\approx$ **0.9998 bits/bit**

Min-Entropy $\approx$ **31.99 bits**

Shannon (Strings) $\approx$ **12.29 bits**

Avg Correlation $\approx$ **0.0115**

Key Finding: Noisy and noiseless configurations produce virtually identical results — the Hadamard + CNOT structure effectively absorbs low-level (2%) noise.

Simulated QRNG vs. ANU Real Quantum Hardware

ANU QRNG generates true random numbers from quantum vacuum fluctuations in a real optical setup. 1024 uint8 values (8192 bits) were fetched for comparison.

Metric	Simulated QRNG (Model D)	ANU Real QRNG	Assessment
Per-Bit Entropy	≈ 0.9999 bits/bit	≈ 0.9997 bits/bit	Both near-ideal ✓
Sequential Correlation	≈ -0.003	$\approx +0.009$	Both near-zero ✓
Source	Qiskit AerSimulator	Real optical hardware	Comparable quality ✓

Comparable Entropy

Both generators achieve per-bit entropy ≥ 0.9997 — near theoretical maximum.

Near-Zero Correlation

Lag-1 autocorrelation close to zero for both — no predictability in sequence.

Simulation Validity

Results suggest the simulation faithfully models real QRNG behaviour at tested metrics.

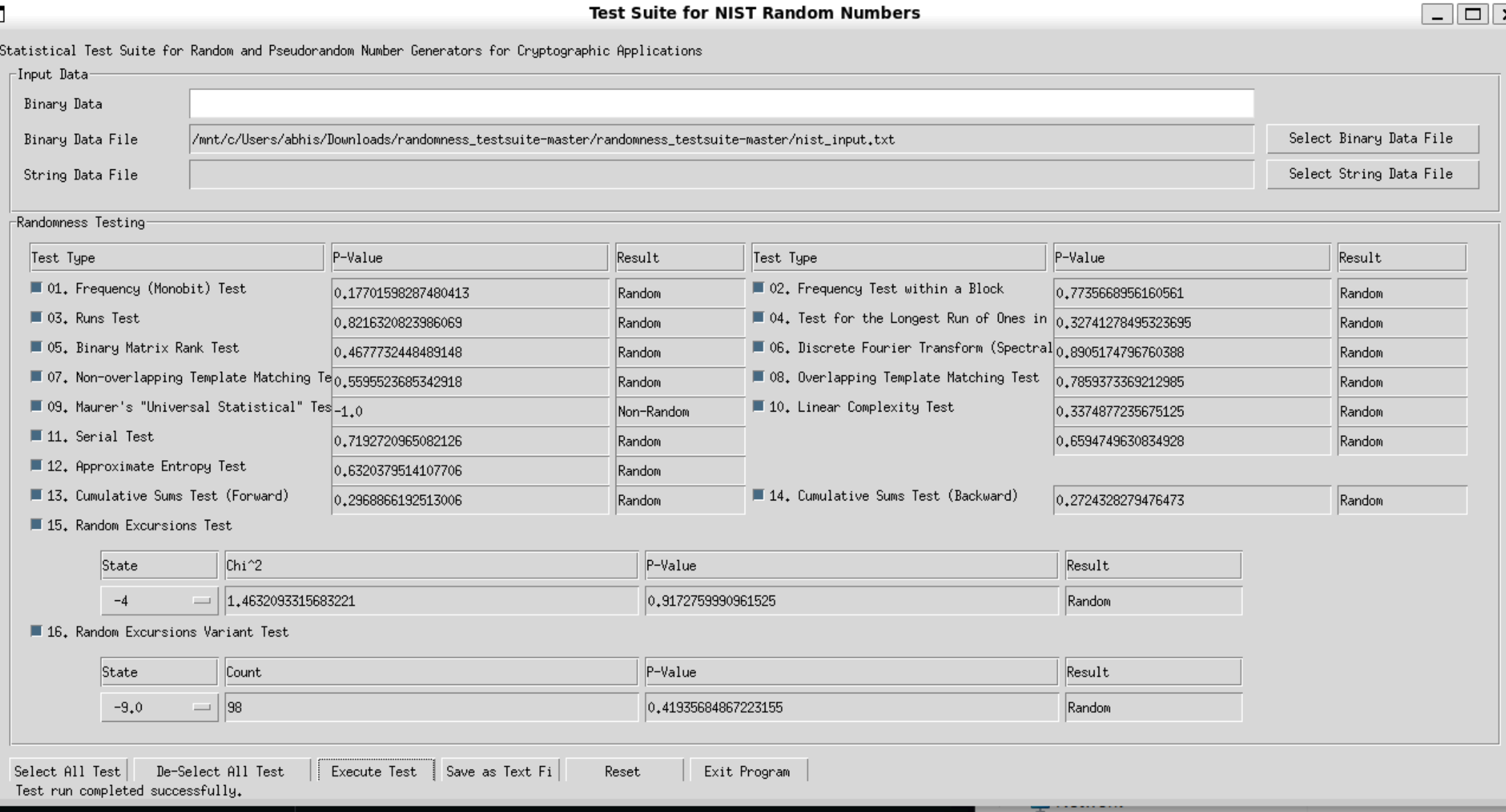
Statistical Randomness Test Suite Results

Test Suite Summary

01. Frequency (Monobit)	Random
02. Freq. within Block	Random
03. Runs Test	Random
04. Longest Run of 1s	Random
05. Binary Matrix Rank	Random
06. DFT (Spectral)	Random
07. Non-overlapping Tmpl	Random
08. Overlapping Template	Random
09. Maurer's Universal	Non-Random⚠
10. Linear Complexity	Random
11. Serial Test	Random
12. Approx. Entropy	Random
13. Cumul. Sums Fwd	Random
14. Cumul. Sums Bwd	Random
15. Random Excursions	Random
16. RE Variant	Random

15/16 PASS

1 NOTE: Test 09



NIST SP 800-22 Test Suite — Actual Run Screenshot

Understanding Test 09: Maurer's Universal Statistical Test

⚠ Test 09 (Maurer's Universal Statistical Test) returned p-value = -1.0 → flagged as Non-Random

What is Maurer's Test?

Measures the compressibility of the bit sequence — a non-compressible sequence is considered random. It requires a minimum bit length (typically 387,840+ bits) to produce meaningful results.

Why $p = -1.0$?

A p-value of exactly -1.0 is a sentinel value indicating the test could not execute properly — most likely due to insufficient bit sequence length. This is a data-size limitation, not a randomness failure.

Is this a problem?

No. This result does not indicate non-random output. It indicates the sample size provided to the test was too small for this specific algorithm. All other 15 tests passed with $p > 0.05$.

What's needed?

To pass Test 09, a much larger output sample is required (hundreds of thousands of bits). This would require significantly more quantum shots or real hardware execution.

Limitations & Future Work

Current Limitations

Noise levels tested (2%) are mild — real hardware error rates can be 5–20%

Independent bit-flip noise model; real noise is correlated across qubit pairs

Per-bit entropy near 1.0 is necessary but not sufficient for full uniformity

ANU comparison limited to only 8,192 bits — too small for statistical rigor

Test 09 (Maurer's) failed due to insufficient sample size, not poor randomness

Future Work

Re-run noise analysis at 5%, 10%, 20% error rates to find degradation threshold

Implement correlated, structured noise models matching real chip topologies

Run full NIST SP 800-22 suite with larger sample (1M+ bits)

Deploy on real IBM Quantum hardware for direct physical validation

Explore qubit count scaling (64, 128-qubit) for higher-bit output quality

Key Takeaways

Model D — Best Architecture

32-qubit Hybrid QRNG with Dice-CNOT + Dice-Measurement achieves max entropy, zero Shannon–Min gap, and lowest correlation (0.0110) at 5000 shots.

Pipeline is Noise-Robust

The five-layer noise pipeline maintains near-maximal entropy and near-zero correlation under 2% init and measurement noise. Hadamard + CNOT absorbs low-level errors.

15/16 NIST Tests Passed

All 15 applicable NIST SP 800-22 tests passed. Test 09 failure is a sample-size limitation (needs 387,840+ bits), not a randomness defect.

Validated Against Real Hardware

Simulated QRNG is statistically comparable to ANU's real quantum hardware at per-bit entropy (≈ 0.9999 vs 0.9997) and autocorrelation metrics.